

Two Library Survey Plans

AP Statistics · Unit 3 · Topic 3.3 · TEACHER KEY

CED TETHER

- **Skill 1.D** — Identify an appropriate inference method for confidence intervals.
- **Skill 4.C** — Verify that inference procedures apply in a given situation.
- **Skill 3.D** — Construct a confidence interval, provided conditions for inference are met.
- **EU UNC-4** — An interval of values should be used to estimate parameters, in order to account for uncertainty.
- **LO UNC-4.A** — Identify an appropriate confidence interval procedure for a population proportion. [Skill 1.D]
- **EK UNC-4.A.1** — The appropriate confidence interval procedure for a one-sample proportion for one categorical variable is a one sample z-interval for a proportion.
- **LO UNC-4.B** — Verify the conditions for calculating confidence intervals for a population proportion. [Skill 4.C]
- **EK UNC-4.B.1** — In order to make assumptions necessary for inference on population proportions, means, and slopes, we must check for independence in data collection methods and for selection of the appropriate sampling distribution.
- **EK UNC-4.B.2** — In order to calculate a confidence interval to estimate a population proportion p , we must check for independence and that the sampling distribution is approximately normal: a. To check for independence: (i) Data should be collected using a random sample or a randomized experiment. (ii) When sampling without replacement, check that $n \leq 10\%N$, where N is the size of the population.
- b. To check that the sampling distribution of $\hat{p}$ is approximately normal (shape): for categorical variables, check that both the number of successes, $n\hat{p}$, and the number of failures, $n(1 - \hat{p})$ are at least 10.
- **LO UNC-4.C** — Determine the margin of error for a given sample size and an estimate for the sample size that will result in a given margin of error for a population proportion. [Skill 3.D]
- **EK UNC-4.C.1** — The standard error of $\hat{p}$ is $SE = \sqrt{\hat{p}(1 - \hat{p})/n}$.
- **EK UNC-4.C.2** — A margin of error gives how much a value of a sample statistic is likely to vary from the value of the corresponding population parameter.
- **EK UNC-4.C.3** — For categorical variables, the margin of error is $z^* \times \sqrt{\hat{p}(1 - \hat{p})/n}$ for a one-sample proportion.
- **EK UNC-4.C.4** — The formula for margin of error can be rearranged to solve for n , the minimum sample size needed to achieve a given margin of error. Use a guess for $\hat{p}$ or use $\hat{p} = 0.5$ to find an upper bound for the sample size.
- **LO UNC-4.D** — Calculate an appropriate confidence interval for a population proportion. [Skill 3.D]
- **EK UNC-4.D.1** — For a one-sample proportion, the interval estimate is $\hat{p} \pm z^* \times \sqrt{\hat{p}(1 - \hat{p})/n}$.
- **EK UNC-4.D.2** — Critical values represent the boundaries encompassing the middle $C\%$ of the standard normal distribution, where $C\%$ is an approximate confidence level for a proportion.
- **LO UNC-4.E** — Calculate an interval estimate based on a confidence interval for a population proportion. [Skill 3.D]
- **EK UNC-4.E.1** — Confidence intervals for population proportions can be used to calculate interval estimates with specified units.

Essential question: Why can a wider interval from a random sample be more trustworthy than a narrower formula result?

DOK-3 skill: 4.C · **FRQ pattern:** narrower-self-selected-interval-vs-wider-random-interval

DOK rationale: Part (c) requires judging two competing interval reports by coordinating recruitment, numerical conditions, and interval width, then explaining why the more precise-looking output cannot support the target-population claim.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- Which condition depends on how cardholders entered the sample?
- What do the numerical conditions not establish?
- Why is Plan B's formula interval narrower?

[WARN] LOOK FOR

- Treating a larger self-selected group as representative.
- Using numerical checks as substitutes for random selection.
- Calling Plan B's output a population confidence interval.

SCORING PART (c) — E / P / I

Expected elements

- **judgment-1** (required) — Chooses A using its SRS and a valid numerical condition.
- **judgment-2** (required) — Compares B width 0.0661 with A width 0.1368.
- **judgment-3** (required) — Explains that B self-selection can bias its estimate and its narrow formula margin does not justify population confidence.

E	Meets all three checks with correct contextual reasoning; equivalent wording is acceptable.
P	Shows at least one substantive correct check, but has an omission or error that prevents E.
I	Shows none of the three checks with substantive correct reasoning; a choice alone is insufficient.

[STAR] THE PROBLEM — annotated

A library system has 20,000 adult cardholders and wants to estimate p , the proportion who used a digital audiobook last month.

Plan A: A computer takes an SRS of 200 cardholder IDs. All answer; 116 say yes.

Plan B: A newsletter link lets cardholders choose to respond once. Of 800 respondents, 520 say yes.

Eli: “Use Plan B. Its larger sample gives the narrower interval.”

Dana: “Use Plan A. Entry into the sample matters more than the narrowest output.”

The 95% formula output for B is (0.6169, 0.6831), width 0.0661. A’s formula width is 0.1368.

Plan counts

Plan	N	n	Yes	No
A	20,000	200	116	84
B	20,000	800	520	280

FIRST TAKE — one sentence before you work the parts

Which plan, if either, should estimate all adult cardholders? Cite one collection detail.

Key: Not graded. The larger sample is intentionally attractive.

[BOOK] ONE-SAMPLE INTERVAL FOR A PROPORTION (keep on the page)

Use a one-sample z -interval for p : $\hat{p} \pm z^* \sqrt{\hat{p}(1-\hat{p})/n}$; for 95% confidence, $z^* = 1.96$. Check (1) random data collection, (2) when sampling without replacement, $n \leq 0.10N$, and (3) at least 10 observed successes and 10 failures. A smaller formula margin of error describes less random sampling variation; it does not repair systematic error caused by who chooses to respond.

DOK 1 (a) Define the population parameter p in context and name the confidence-interval procedure that would be appropriate when its conditions are met.

Key: Here p is the proportion of all 20,000 adult library cardholders who used a digital audiobook last month. When conditions are met, the procedure is a one-sample z -interval for a population proportion.

DOK 2 (b) For Plan A, calculate $\hat{p}$ and its 95% interval. Check A’s random, 10 percent, and success/failure conditions. For Plan B, identify the condition that fails.

Key: Plan A has $\hat{p} = 116/200 = 0.58$. It is an SRS; $200 \leq 0.10(20,000) = 2,000$; and counts are 116 and 84. Thus $SE = 0.0348999$, $ME = 1.96(SE) = 0.0684037$, and the valid interval is (0.5116, 0.6484). Plan B has $\hat{p} = 520/800 = 0.65$; $800 \leq 2,000$ and counts 520 and 280 pass, but self-selection fails the random condition. Its mechanical output is 0.65 ± 0.0330523 , or (0.6169, 0.6831), not a defensible population confidence interval.

DOK 3 (c) In 3–4 sentences, choose the plan that can estimate all cardholders. Use recruitment and one numerical condition. Explain why the supplied widths do or do not support Eli’s recommendation.

Key: Use Plan A. Its SRS, $200 \leq 2,000$, and counts 116 and 84 support the interval (0.512, 0.648) for all cardholders. Its width is about 0.1368 versus Plan B’s smaller formula width, 0.0661. Plan B’s volunteers may differ systematically from nonrespondents; its margin of error covers random variation, not self-selection. Reporting it would falsely suggest that a narrow range represents all cardholders with 95 percent confidence.